ADITYA AYUSHMAN SAHOO

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B.Tech, Computer Science and Engineering - Artificial Intelligence and Machine Learning
Address : SRM INSTITUTE OF SCIENCE AND TECHNOLOGY KTR CHENNAI

Summary

AI and Machine Learning undergraduate with strong interests in Computer Vision, Deep Learning, Cloud Computing, and Intelligent Systems. Experienced in developing realtime AI analytics platforms, cloud optimization systems, and image processing applications using modern machine learning frameworks and scalable backend technologies. Passionate about building impactful AI-driven solutions for real-world problems.

Links

LinkedIn, GitHub, LeetCode

Education

SRM IST-Kattankulathur 2027
B.Tech · Computer Science and Engineering - Artificial Intelligence and Machine Learning CGPA - 7.46/10

DAV PUBLIC SCHOOL CSPUR BBSR 2023
Class XII - CBSE · BHUBANESWAR ODISHA Percentage - 65.6%

DAV PUBLIC SCHOOL CSPUR BBSR 2021
Class X - CBSE · BHUBANESWAR ODISHA Percentage - 90.4%

Experience

Osswal Infosystem Pvt. Ltd. | SAP Partner Jul 2026 - Present
Internship · Flutter Developer Intern Indore

- Developing cross-platform mobile app features in Flutter (Dart) for a client project built on SAP Business One
- Learning enterprise mobile development alongside the team, from building UI screens to connecting them with SAP Business One business data and workflows
- Working with senior developers and SAP consultants through the development cycle, including requirements, implementation, testing and code reviews

KIIT - Kalinga Institute of Industrial Technology Jun 2026 - Present
Research internship · Research Intern

- Researching AI-based medical image analysis with segmentation, classification and explainable AI for clinical decision support (MedVision, see Projects)
- Running experiments on segmentation and explainable-AI methods and documenting results for the research group

Kukreja's Wastec Bio-Gas Pvt. Ltd Jun 2026 - Present
Internship · Intern Mumbai

- Owned development of an AI-based visual inspection system (PCBMind AI, detailed in Projects) for the company's electronics QC automation initiative, covering control boards used in plant equipment
- Gathered inspection requirements from the team, defined the defect categories and severity levels, and delivered a working web dashboard for QC review and reporting

Syllogistek Systems Private Ltd. Jun 2025 - Jul 2025
· Summer Intern · Internship

- Developed AI-based software solutions using Generative AI, Keras and PyTorch
- Worked with technical teams to streamline implementation workflows and improve system processes

Projects

PCBMind AI — Automated PCB Defect Inspection | Live Demo
Jun 2026 - Present
Kukreja's Wastec Bio-Gas Pvt. Ltd. · Python, PyTorch, YOLOv8, OpenCV, FastAPI, React.js, PostgreSQL, Supabase, Docker, Git
Computer Vision / Manufacturing Quality Control

Built an AI system that automates defect inspection of printed circuit boards for electronics manufacturing. It aligns each board image against a reference (CLAHE, ORB feature matching, homography), detects defects with YOLOv8 and classifies how severe they are. A FastAPI backend and React dashboard show inspection analytics and generate annotated PDF/CSV quality reports automatically.

CrowdCount — Multi-Zone Real-Time Crowd Intelligence System
Feb 2026 - May 2026
Academic Project – SRMIST · Python, YOLOv8, OpenCV, FastAPI
Computer Vision / Smart Infrastructure

Developed a real-time crowd monitoring platform that uses YOLOv8 to detect people and track crowd levels across several zones. Features include centroid tracking, heatmaps, line-crossing counts and region-of-interest analytics. It handles both live camera feeds and recorded video, with a FastAPI backend for real-time metrics, alerts and structured reports.

GridSense — Carbon-Aware EV Charging Optimizer | Live Demo
Sept 2026 - Present
· Python, PuLP, FastAPI, PostgreSQL, React, TypeScript, Docker, Vercel, Render
Energy Systems / Optimization

Built a platform to reduce the grid strain and carbon emissions caused by unmanaged home EV charging. A linear-programming optimizer schedules charging into off-peak, low-carbon hours without missing each user's deadline. It's a full-stack system with a FastAPI backend, PostgreSQL storage and a React dashboard, containerized with Docker and deployed through CI/CD.

Cloud Computing Resource Optimizer | SSRN Publication
Jan 2026 - May 2026
Academic Research – SRMIST (published on SSRN) · Python, XGBoost, Random Forest, Deep Q-Network (DQN), Explainable AI
Machine Learning / Reinforcement Learning / Cloud Computing

Built a system that decides how much computing power to give cloud workloads. XGBoost and Random Forest models forecast CPU and memory demand, and a Deep Q-Network scheduler adjusts resource allocation as demand changes. Explainable AI makes each allocation decision easy to interpret. The work is available as a research paper on SSRN (2026).

MedVision — AI-Powered Medical Imaging Intelligence | Live Demo
Jun 2026 - Present
KIIT – Kalinga Institute of Industrial Technology (Research Internship) · Python, PyTorch, OpenCV, U-Net, ResNet, VGG16, Grad-CAM, FastAPI, React.js, PostgreSQL
Computer Vision / Deep Learning / Explainable AI

Developing an AI system that helps doctors make clinical decisions by analysing medical scans. The pipeline covers DICOM image processing, U-Net segmentation to locate problem regions, and ResNet/VGG16 transfer learning for classification. Grad-CAM heatmaps show which part of the scan drove each prediction, so doctors can see why the model decided what it did. The system also tracks how a patient's condition changes across repeated scans.

Smart Road Guardian AI X — Edge AI Road Safety Platform | Live Demo
Jun 2026 - Sept 2026
· Python, PyTorch, ONNX, TensorRT, pgmpy, NVIDIA Jetson, Next.js, Supabase
Edge AI / Causal Machine Learning / Transportation Safety

Developed an edge AI platform on NVIDIA Jetson that predicts and helps prevent road accidents. It uses causal Bayesian networks to find the actual causes of risk rather than just correlations. An MVP is built with live monitoring, fleet and driver management, an alerts center and what-if risk analysis.

Skills

Programming Languages: Java, C, Python, SQL, JavaScript, C++, TypeScript
Technical Skills: Deep Learning, Machine Learning, Computer Vision, Reinforcement Learning, Deep Q Network, DBMS, Operating Systems, Computer Networks, Data Structures & Algorithms
Languages and Frameworks: NumPy, Matplotlib, YOLOv8, FastAPI, React.js, OpenCV, TensorFlow, XGBoost, PyTorch, Flutter
Tools and Technologies: PostgreSQL, Docker, Git, Vercel, Render, Power BI, SAP Business One
Hardware Skills: Arduino, IoT Sensor Integration
Domain Knowledge: Manufacturing Quality Inspection, Energy Optimization, Explainable AI
Areas of Interest: Edge AI, Smart Infrastructure, Industrial Automation

Publications

Cloud Computing Resource Optimizer | SSRN
Jul 2026
SSRN

Competitions

Samsung ennovateX AX Hackathon 2026
· Team OMNIRAN NEXUS
Built a multimodal agentic RAG assistant for telecom Radio Access Network operations, combining cross-modal retrieval with agentic reasoning. Also submitted AURA, an AI tender evaluation and fraud detection platform, under Theme 3.

Google Solution Challenge 2026 Jul 2026
Took part in Google's global student developer challenge, designing a solution that uses Google technologies to address a UN Sustainable Development Goal.

Adobe University Hackathon Aug 2026
Competed in Adobe's national university hackathon, working in a team to design and prototype a technical solution to a real-world problem under strict time constraints.

Hack the Cosmos'25 May 2025
Participated in a competitive hackathon, collaborating with a team to ideate, build and pitch a working prototype within the event timeline.

Certifications

NPTEL Certified Programming in Java | Link Jul 2024
NPTEL

Networking Basics | Link Nov 2025
Cisco

Deep Learning Onramp | Link Apr 2025
MathWorks

Tata - Data Visualisation: Empowering Business with Effective Insights Job Simulation | Link Jun 2026
Forage · SYu6aXzmDCcS9Xf7F

Cybersecurity Virtual Internship | Link Jun 2026
AICTE NEAT · 4a1b9aea7f3b03947329

Extra Curricular Activities

Committee Member, Aaruush Aug 2023 - Aug 2024

Committee Head, Aaruush Aug 2024

Member, SRM MUN Society Oct 2023

Languages

ENGLISH [Professional Working Proficiency], ODIA [Native Proficiency], HINDI [Professional Working Proficiency], SPANISH [Fundamental Proficiency]

Volunteering

BookMyShow Feb 2025
Event Volunteer · Ed Sheeran India Tour 2025

- Selected as an official volunteer for the Ed Sheeran India Tour 2025 event operations in Chennai
- Assisted in crowd coordination, audience management, and large-scale event execution activities